MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TRANSFORMER, SIGNAL, ANALOG, KEY CODE CGR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic signal transformer module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.
MIL-T-27	-	Transformers and Inductors (Audio, Power, and High Power Pulse), General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein and the transformers shall comply with the applicable requirements of MIL-T-27.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.05 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be CGR.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A20 and B21 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/240-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

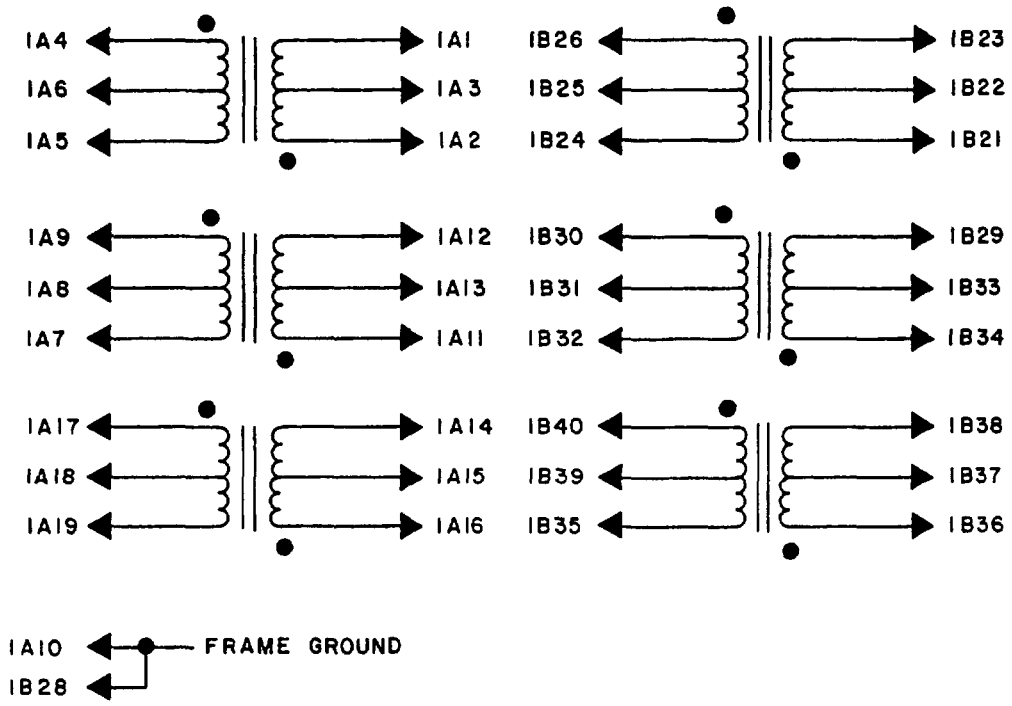
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. Unused pins are 1A20, 1B27.
2. Physically adjacent transformers shall be oriented at 90° to each other.

FIGURE 1. CGR module function diagram, signal transformer.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A6	Isolation, primary to secondary	II	X	20	∞	10	∞	20	∞	10	∞	MΩ
A7-A12	Isolation, between transformers	II	S	20	∞	10	∞	20	∞	10	∞	MΩ
A13	Isolation, frame to pins	II	S	20	∞	10	∞	20	∞	10	∞	MΩ
A14	Isolation, used to unused pins	II	X	20	∞	10	∞	20	∞	10	∞	MΩ
A15	Isolation, between unused pins	II	X	20	∞	10	∞	20	∞	10	∞	MΩ
B1-B8	Dielectric withstanding voltage check	III	S	0	30	0	60	0	30	0	60	μA
C1-C6	Transformation ratio check, full primary to full secondary	IV	X	0.869	1.010	0.869	1.010	0.860	1.010	0.850	1.010	Vrms
C7-C18	Transformation ratio check	IV	X	← Table IV limits →								
D1-D6	Phase shift, 200 Hz	V	X	180.0	186.0	180.0	186.0	180.5	186.8	180.5	186.8	Degree
D7-D12	Phase shift, 5 kHz	V	X	178.4	180.0	178.4	180.0	178.4	180.0	178.4	180.0	Degree
D13-D18	Phase shift, 40 kHz	V	X	169	172	169	172	169	172	169	172	Degree
D19-D24	Phase shift, 500 Hz	V	X	180.0	182.5	180.0	182.5	180.0	183.2	180.0	183.2	Degree
E1-E6	DC resistance, primary	VI	S	0	103	0	103	0	126	0	128	Ω
E7-E12	DC resistance, secondary	VI	S	0	129	0	129	0	158	0	160	Ω
F1-F12	Insertion loss check	VII	X	0.869	1.010	0.869	1.010	0.860	1.010	0.850	1.010	Vrms
G1-G6	Frequency response check, 5 kHz	VIII	X	0.869	1.010	0.869	1.010	0.860	1.010	0.850	1.010	Vrms
G7-G18, G25-G30	Frequency response check	VIII	X	← Table VIII limits →								
G19-G24, G31-G48	Frequency response check	VIII	S	← Table VIII limits →								
H1-H6	Primary inductance (1 kHz)	IX	S	910	∞	910	∞	910	∞	910	∞	mH

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J1-J18	Maximum power check (distortion test)	X	S	0	5	0	5	0	5	0	5	%
K1-K36	Magnetic shielding check	XI	Q	-∞	-80	-∞	-80	-∞	-80	-∞	-80	dBV
L1-L45	Output crosstalk	XII	S	-∞	-70	-∞	-70	-∞	-70	-∞	-70	dBV
M1,M2	Continuity	XIII	X	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through M shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through M in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A1 through A6, C through J, and M1 shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions A1 through A6 shall not be performed at 0°C. Test conditions A1 through A6, A14, A15, and M2 shall not be performed during the transient thermal test. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be nonoperating. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through J and M1 shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through J, M1, and M2 shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

* 4.5.4 Stress test. The module shall be operated as shown on figure 3. Following the stress test, electrical tests A through J, M1, and M2 shall be performed in any order using the 25°C ±5°C ambient initial requirements of table I. The stress test shall be performed on the module subjected to vibration prior to the vibration test.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331 and MIL-T-27.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/240-1 supersedes part number M28787/240. The two parts are interchangeable.

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Individual transformer description:

Dielectric withstanding voltage	300 V_{p-p}(ac)
Maximum input voltage	±16.5 V dc (abnormal) 8.5 Vrms at 1 kHz (normal)
Primary to secondary turns ratio	1:1
Center tap accuracy	
Primary	Within ±1% of primary electrical center.
Secondary	Within ±1% of secondary electrical center.
Phase shift	185.0° $^{+1.8°}_{-5.0°}$ at low frequencies 179° $^{+1.0°}_{-0.6°}$ at mid frequencies (approximately 5 kHz). 171° $^{+1°}_{-2°}$ at high frequencies (output lagging input)
Typical impedance	
Primary and secondary	1,000Ω nominal (not measured)
Insertion loss (500 Hz to 10 kHz)	1.4 dB maximum with 2 kΩ load and low source impedance
Frequency response	
500 Hz to 10 kHz	±0.5 dB of response at 5 kHz
50 Hz ≤ f < 500 Hz	+0.5 dB, -3.0 dB of response at 5 kHz
10 kHz < f ≤ 300 kHz	+0.5 dB, -4.0 dB of response at 5 kHz
Maximum power (with unbalanced primary current of 2.5 mA)	0.5 mW at 200 Hz, 10 mW at 400 Hz, and 35 mW at 1,000 Hz into 2 kΩ with ≤ 5% total harmonic distortion
DC unbalanced current	2.5 mA maximum
Isolation	
Between primary and secondary	≥ 10 MΩ at 500 V dc
Between transformers	≥ 10 MΩ at 500 V dc
Output crosstalk	< -70 dB from input of each transformer to output of all other transformers.

* 6.5 Module function. This module contains six signal transformers which provide analog signal coupling.

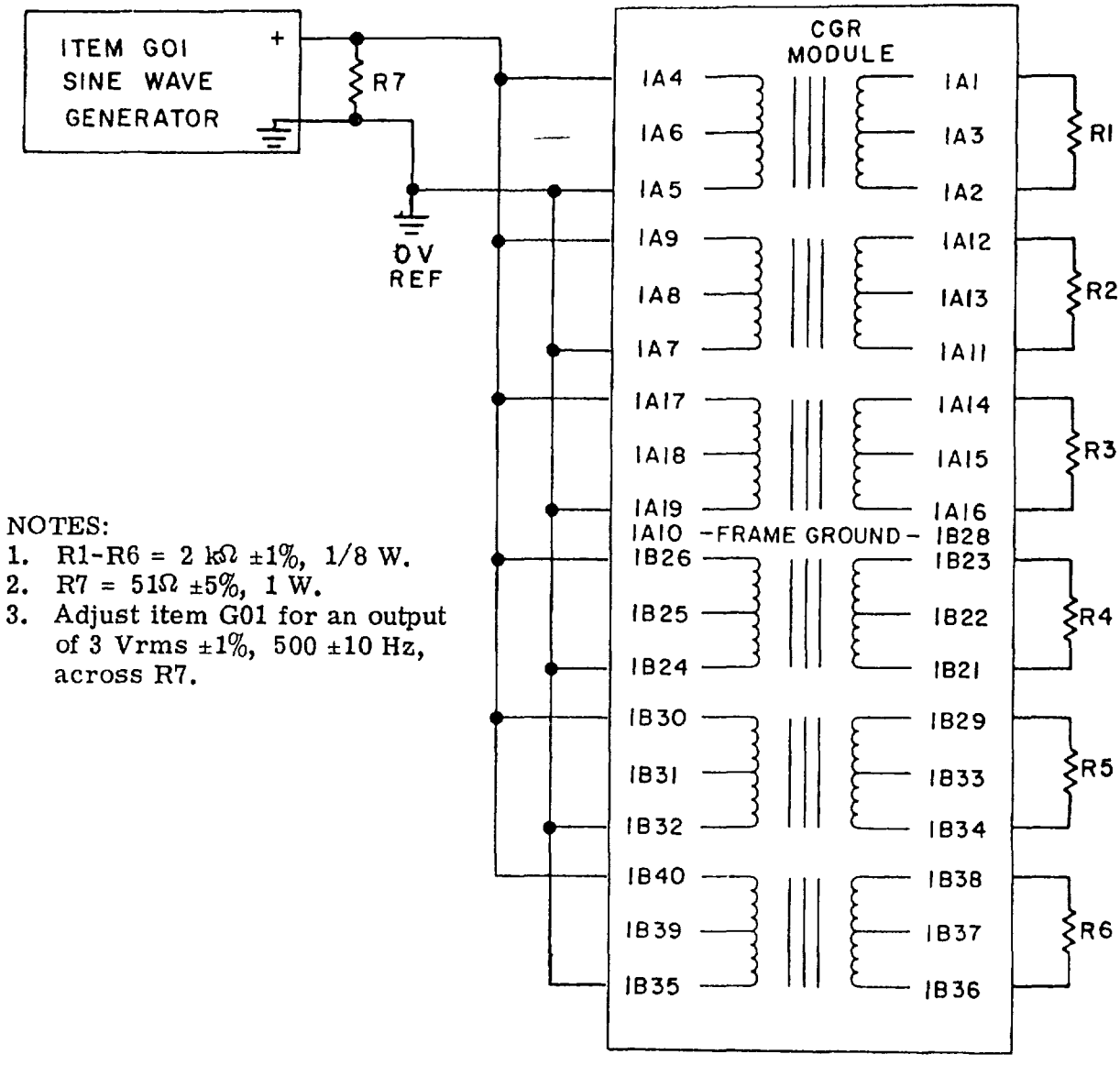
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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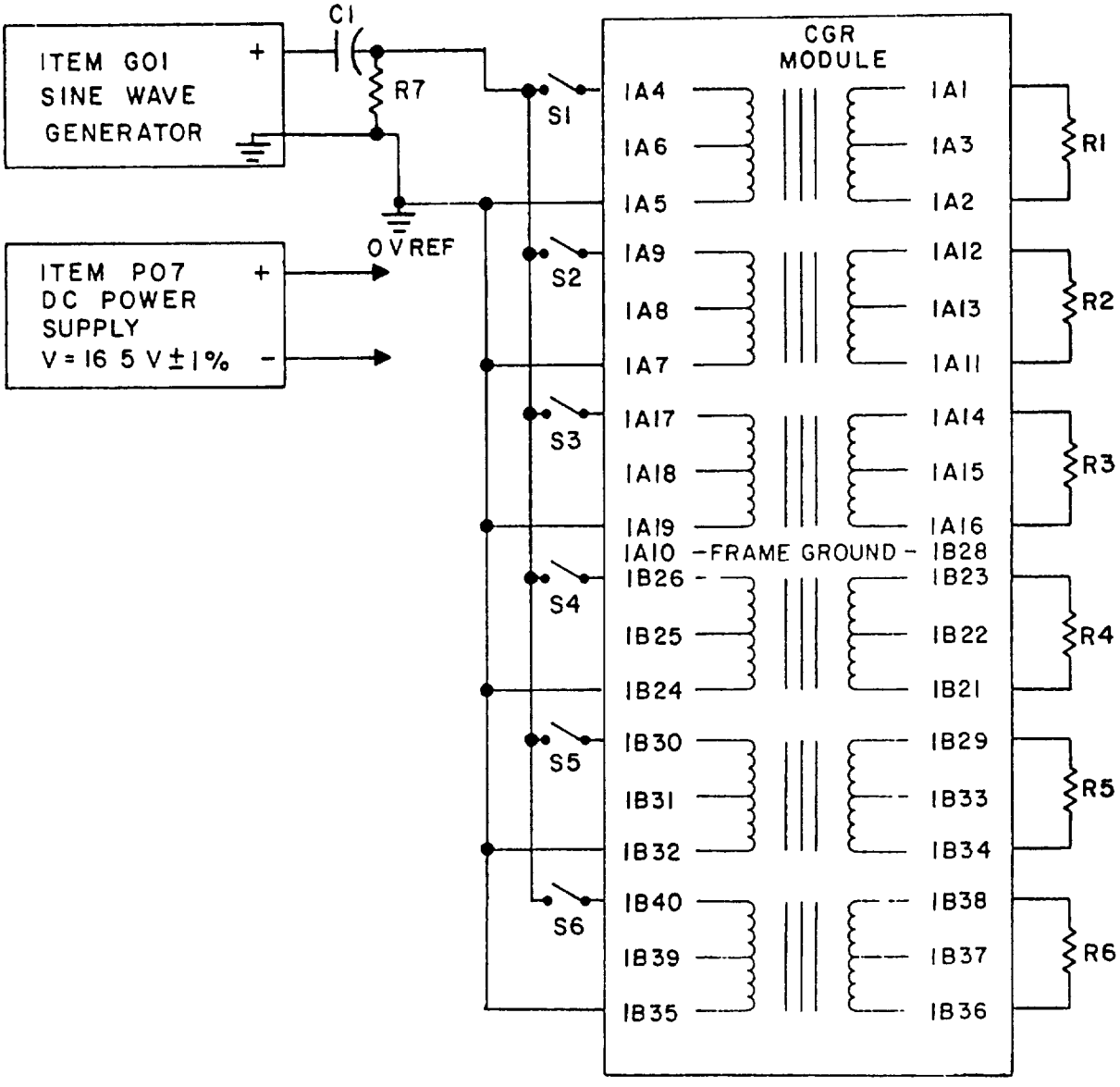
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NOTES:
1. R1-R6 = 2 kΩ ±1%, 1/8 W.
2. R7 = 51Ω ±5%, 1 W.
3. Adjust item G01 for an output of 3 Vrms ±1%, 500 ±10 Hz, across R7.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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NOTES:
1. R1-R6 = 2 kΩ ±1%, 1/8 W.
2. R7 = 51Ω ±5%, 6 W, wirewound.
3. C1 = 22 µF ±10%, 20 V dc, nonpolarized, tantalum.
4. S1-S6 = SPST switch.
5. Adjust item G01 for an output of 3 Vrms ±1%, 500 ±10 Hz, across R7.
6. Procedure for stress test:

   a. Close switch S1.
   b. Connect item P07 across R7 and leave connected for 5 minutes. Reverse the polarity across R7 and leave connected for an additional 5 minutes.
   c. Open switch S1.
   d. Repeat a through c above for switches S2 through S6. Close only one switch at a time.
   e. Perform table I tests as specified in 4.5.4.

FIGURE 3. Stress test circuit.

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TABLE II. Isolation test.

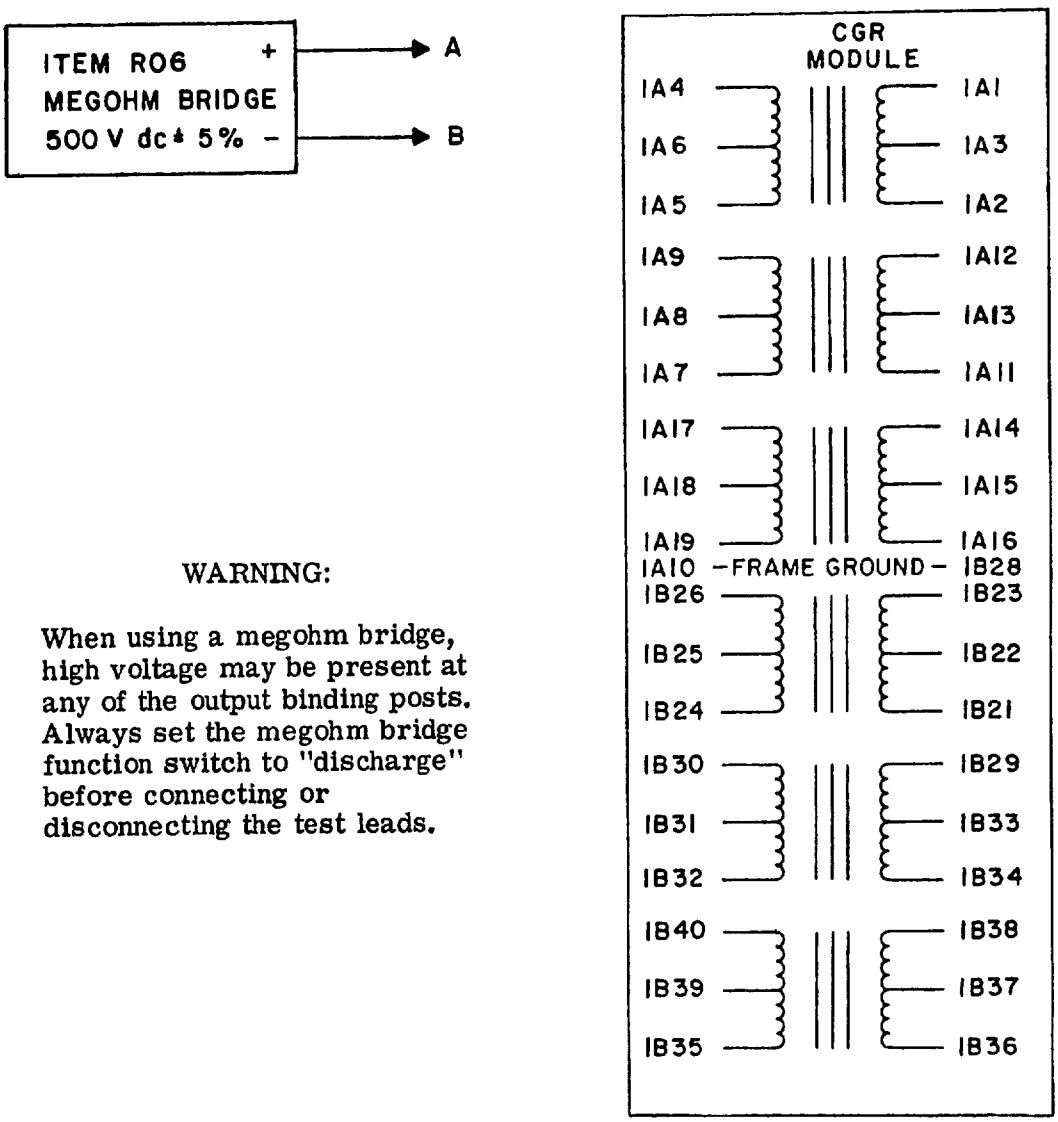
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, primary to secondary	4	1	20	∞	MΩ	4
A2	"	9	12	"	"	"	"
A3	"	17	14	"	"	"	"
A4	"	26	23	"	"	"	"
A5	"	30	29	"	"	"	"
A6	"	40	38	"	"	"	"
A7	Isolation, between transformers	4	12,14,23,29, 38	"	"	"	"
A8	"	9	1,14,23,29,38	"	"	"	"
A9	"	17	1,12,23,29,38	"	"	"	"
A10	"	26	1,12,14,29,38	"	"	"	"
A11	"	30	1,12,14,23,38	"	"	"	"
A12	"	40	1,12,14,23,29	"	"	"	"
A13	Isolation, frame to pins	1,2,3,4,5,6,7, 8,9,11,12,13, 14,15,16,17, 18,19,20,21, 22,23,24,25, 26,27,29,30, 31,32,33,34, 35,36,37,38, 39,40	10,28	"	"	"	"

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TABLE II. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A14	Isolation, used to unused pins	1,2,3,4,5,6,7, 8,9,11,12,13, 14,15,16,17, 18,19,21,22, 23,24,25,26, 29,30,31,32, 33,34,35,36, 37,38,39,40	20,27	20	∞	MΩ	4
A15	Isolation, between unused pins	20	27	20	∞	MΩ	4

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

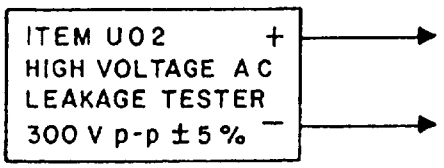
FIGURE 4. Isolation test circuit for table II.

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TABLE III. Dielectric withstanding voltage check test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Dielectric withstanding voltage check	1,4,9,12,14, 17,23,26,29, 30,38,40	10,28	0	30	µA	5
B2	"	1,9,12,14,17, 23,26,29,30, 38,40	4	"	"	"	"
B3	"	1,12,14,17, 23,26,29,30, 38,40	9	"	"	"	"
B4	"	1,12,14,23, 26,29,30,38, 40	17	"	"	"	"
B5	"	1,12,14,23, 29,30,38,40	26	"	"	"	"
B6	"	1,12,14,23, 29,38,40	30	"	"	"	"
B7	"	1,12,14,23, 29,38	40	"	"	"	"
B8	"	10,28	20,27	"	"	"	"

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WARNING:

Danger - High voltage
- Exercise extreme caution.

NOTES:
1. This test is to determine dielectric withstanding voltage. Source level should be adjusted until specified withstanding level is reached or breakdown occurs. Do not exceed 300 V_{p-p}.
2. Before inserting the module into the test fixture, measure the test fixture leakage current at 300 V_{p-p} for each test condition. Subtract this value from the measured leakage current before comparing to the table III limits.

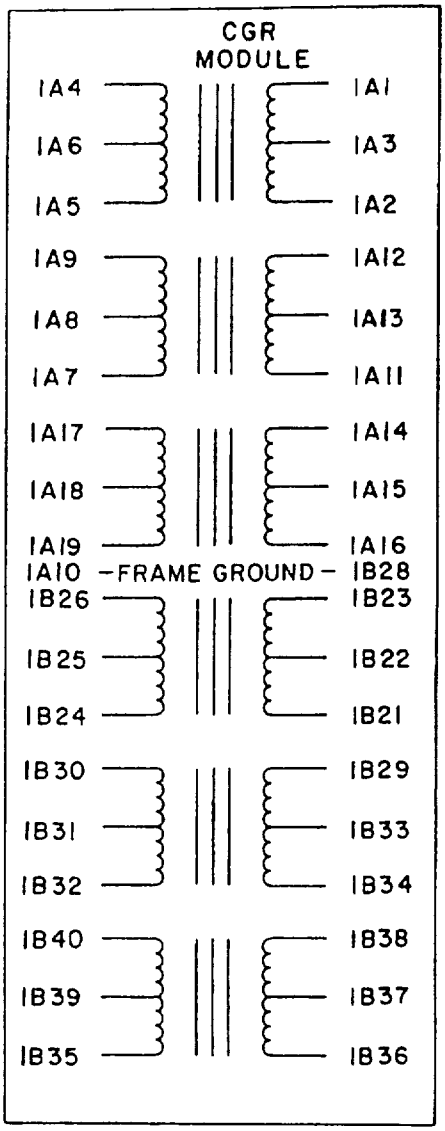


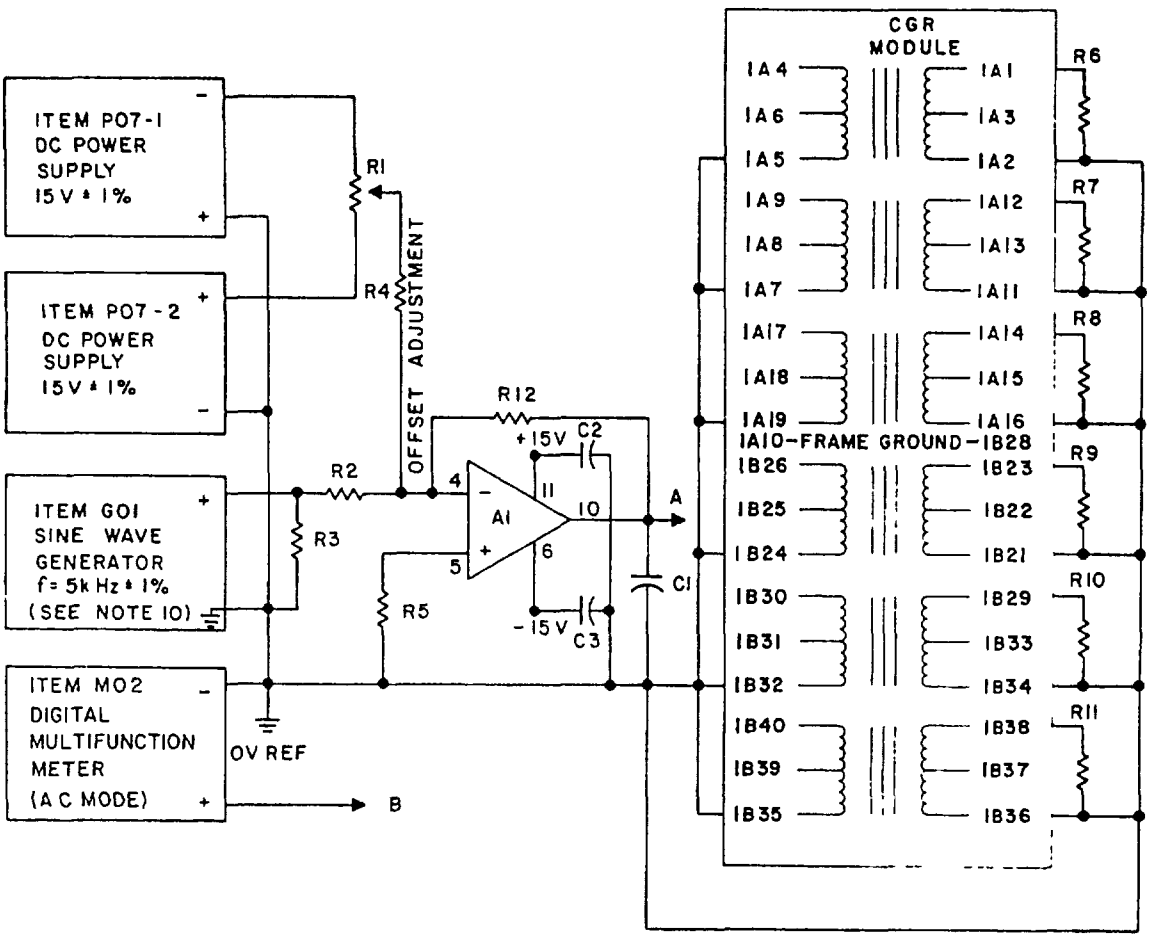
FIGURE 5. Dielectric withstanding voltage check test circuit for table III.

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TABLE IV. Transformation ratio check test.

Test condition	Parameter	Record item M02 reading	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	Transformation ratio check, full primary to full secondary	V1 =	4	1	0.869	1.010	Vrms	6
C2	"	V2 =	9	12	"	"	"	"
C3	"	V3 =	17	14	"	"	"	"
C4	"	V4 =	26	23	"	"	"	"
C5	"	V5 =	30	29	"	"	"	"
C6	"	V6 =	40	38	"	"	"	"
C7	Transformation ratio check, CT primary to full secondary	- - -	6	1	1.875V1	2.012V1	"	"
C8	"	"	8	12	1.875V2	2.012V2	"	"
C9	"	"	18	14	1.875V3	2.012V3	"	"
C10	"	"	25	23	1.875V4	2.012V4	"	"
C11	"	"	31	29	1.875V5	2.012V5	"	"
C12	"	"	39	38	1.875V6	2.012V6	"	"
C13	Transformation ratio check, full primary to CT secondary	"	4	3	0.488V1	0.512V1	"	"
C14	"	"	9	13	0.488V2	0.512V2	"	"
C15	"	"	17	15	0.488V3	0.512V3	"	"
C16	"	"	26	22	0.488V4	0.512V4	"	"
C17	"	"	30	33	0.488V5	0.512V5	"	"
C18	"	"	40	37	0.488V6	0.512V6	"	"

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NOTES:
1. R1 = 10 kΩ ±1%, multi-turn, 1/4 W potentiometer.
2. R2 = R12 = 10 kΩ ±1%, 1/4 W.
3. R3 = 51Ω ±5%, 1 W.
4. R4 = 200 kΩ ±5%, 1/8 W.
5. R5 = 4.99 kΩ ±1%, 1/8 W.
6. R6-R11 = 2 kΩ ±0.1%, 1/8 W.
7. C1 = 300 pF ±10%.
8. C2 = C3 = 0.1 µF ±20%.
9. A1 = µA741 operational amplifier.
10. Adjust R1 for minimum offset out of A1 with item G01 and A1 disconnected from module under test. Use item M02 to measure A1 offset.
11. Adjust item G01 for 1 Vrms ±0.1% output at A for test conditions C1-C6, 1 Vrms ±0.1% for test conditions C7-C12; and 1 Vrms ±0.1% for test conditions C13-C18.
12. Record in the blanks provided in table IV the actual item M02 readings of V1, V2, V3, V4, V5, and V6 for test conditions C1-C6. Use these values in computing the limits for test conditions C7-C18. Compare the actual item M02 readings to computed limits for test conditions C7-C18.

* FIGURE 6. Transformation ratio check test circuit for table IV.

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* TABLE V. Phase shift test.

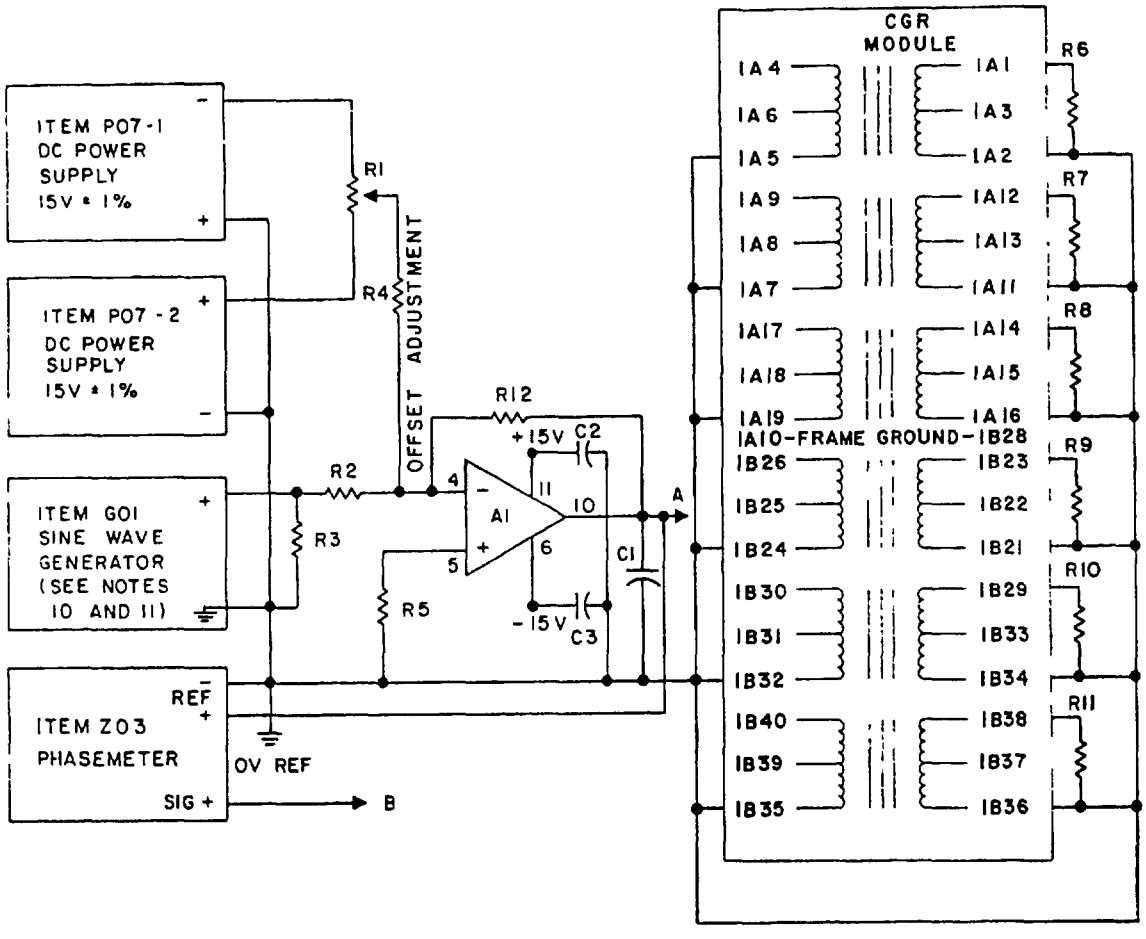
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Phase shift, 200 Hz	4	1	180.0	186.0	Degree	7
D2	"	9	12	"	"	"	"
D3	"	17	14	"	"	"	"
D4	"	26	23	"	"	"	"
D5	"	30	29	"	"	"	"
D6	"	40	38	"	"	"	"
D7	Phase shift, 5 kHz	4	1	178.4	180.0	"	"
D8	"	9	12	"	"	"	"
D9	"	17	14	"	"	"	"
D10	"	26	23	"	"	"	"
D11	"	30	29	"	"	"	"
D12	"	40	38	"	"	"	"
D13	Phase shift, 40 kHz	4	1	169	172	"	"
D14	"	9	12	"	"	"	"
D15	"	17	14	"	"	"	"
D16	"	26	23	"	"	"	"
D17	"	30	29	"	"	"	"
D18	"	40	38	"	"	"	"

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* TABLE V. Phase shift test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D19	Phase shift, 500 Hz	4	1	180.0	182.5	Degree	7
D20	"	9	12	"	"	"	"
D21	"	17	14	"	"	"	"
D22	"	26	23	"	"	"	"
D23	"	30	29	"	"	"	"
D24	"	40	38	"	"	"	"

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NOTES:
1. R1 = 10 kΩ ±1%, multi-turn, 1/4 W, potentiometer.
2. R2 = R12 = 10 kΩ ±1%, 1/4 W.
3. R3 = 51Ω ±5%, 1 W.
4. R4 = 200 kΩ ±5%, 1/8 W.
5. R5 = 4.99 kΩ ±1%, 1/8 W.
6. R6-R11 = 2 kΩ ±1%, 1/8 W.
7. C1 = 300 pF ±10%.
8. C2 = C3 = 0.1 µF ±20%.
9. A1 = µA741 operational amplifier.
10. Adjust R1 for minimum offset out of A1 with item G01 and A1 disconnected from module under test. Use item M02 to measure A1 offset.
11. Adjust item G01 for 50 mVrms ±1% output at A.
12. Adjust item G01 to 200 Hz ±1% for test conditions D1-D6; 5 kHz ±1% for test conditions D7-D12; 40 kHz ±1% for test conditions D13-D18; and 500 Hz ±1% for test conditions D19-D24.
13. The capacitance at point B (including stray, phasemeter, and meter leads) shall be kept to a minimum and shall not exceed 80 pF.

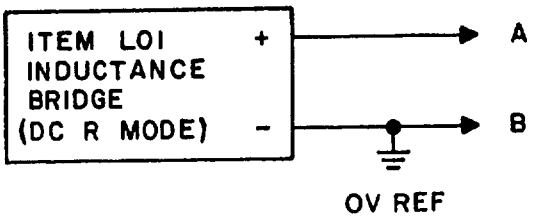
* FIGURE 7. Phase shift test circuit for table V.

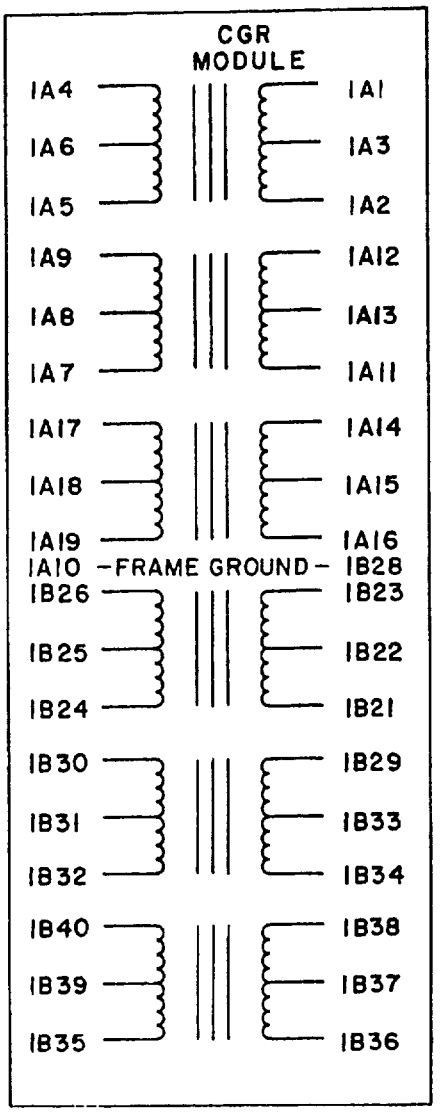
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TABLE VI. DC resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	DC resistance, primary	4	5,7,19,24, 32,35	0	103	Ω	8
E2	"	9	"	"	"	"	"
E3	"	17	"	"	"	"	"
E4	"	26	"	"	"	"	"
E5	"	30	"	"	"	"	"
E6	"	40	"	"	"	"	"
E7	DC resistance, secondary	1	2,11,16,21, 34,36	"	129	"	"
E8	"	12	"	"	"	"	"
E9	"	14	"	"	"	"	"
E10	"	23	"	"	"	"	"
E11	"	29	"	"	"	"	"
E12	"	38	"	"	"	"	"

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NOTE: Adjust item LOI for 0.1 V dc across transformer.

FIGURE 8. DC resistance test circuit for table VI.

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TABLE VII. Insertion loss check test.

Test condition	Parameter	S1 position	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
F1	Insertion loss check, 500 Hz	1	0.869	1.010	Vrms	9
F2	"	2	"	"	"	"
F3	"	3	"	"	"	"
F4	"	4	"	"	"	"
F5	"	5	"	"	"	"
F6	"	6	"	"	"	"
F7	Insertion loss check, 10 kHz	6	"	"	"	"
F8	"	5	"	"	"	"
F9	"	4	"	"	"	"
F10	"	3	"	"	"	"
F11	"	2	"	"	"	"
F12	"	1	"	"	"	"

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* TABLE VIII. Frequency response check test.

Test condition	Parameter	Record item M02 reading 1/	S1 position	Item G01 frequency ±1%	Initial limits 25°C ±5°C ambient		Unit	Test figure
					Min	Max		
G1	Frequency response check	V1' =	1	5 kHz	0.869	1.010	Vrms	9
G2	"	V2 =	2	"	"	"	"	"
G3	"	V3 =	3	"	"	"	"	"
G4	"	V4 =	4	"	"	"	"	"
G5	"	V5 =	5	"	"	"	"	"
G6	"	V6 =	6	"	"	"	"	"
G7	"	- - -	1	50 Hz	V1' -3	V1' +0.5	dBV	"
G8	"	"	2	"	V2' -3	V2' +0.5	"	"
G9	"	"	3	"	V3' -3	V3' +0.5	"	"
G10	"	"	4	"	V4' -3	V4' +0.5	"	"
G11	"	"	5	"	V5' -3	V5' +0.5	"	"
G12	"	"	6	"	V6' -3	V6' +0.5	"	"
G13	"	"	1	500 Hz	V1' -0.5	V1' +0.5	"	"
G14	"	"	2	"	V2' -0.5	V2' +0.5	"	"
G15	"	"	3	"	V3' -0.5	V3' +0.5	"	"
G16	"	"	4	"	V4' -0.5	V4' +0.5	"	"
G17	"	"	5	"	V5' -0.5	V5' +0.5	"	"
G18	"	"	6	"	V6' -0.5	V6' +0.5	"	"

See footnote at end of table.

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* TABLE VIII. Frequency response check test - Continued.

Test condition	Parameter	Record item M02 reading 1/	S1 position	Item G01 frequency ±1%	Initial limits 25°C ±5°C ambient		Unit	Test figure
					Min	Max		
G19	Frequency response check	- - -	1	1 kHz	V1' -0.5	V1' +0.5	dBV	9
G20	"	"	2	"	V2' -0.5	V2' +0.5	"	"
G21	"	"	3	"	V3' -0.5	V3' +0.5	"	"
G22	"	"	4	"	V4' -0.5	V4' +0.5	"	"
G23	"	"	5	"	V5' -0.5	V5' +0.5	"	"
G24	"	"	6	"	V6' -0.5	V6' +0.5	"	"
G25	"	"	1	10 kHz	V1' -0.5	V1' +0.5	"	"
G26	"	"	2	"	V2' -0.5	V2' +0.5	"	"
G27	"	"	3	"	V3' -0.5	V3' +0.5	"	"
G28	"	"	4	"	V4' -0.5	V4' +0.5	"	"
G29	"	"	5	"	V5' -0.5	V5' +0.5	"	"
G30	"	"	6	"	V6' -0.5	V6' +0.5	"	"
G31	"	"	1	400 kHz	V7' -6	V7' +0.5	"	"
G32	"	"	2	"	V8' -6	V8' +0.5	"	"
G33	"	"	3	"	V9' -6	V9' +0.5	"	"
G34	"	"	4	"	V10' -6	V10' +0.5	"	"
G35	"	"	5	"	V11' -6	V11' +0.5	"	"
G36	"	"	6	"	V12' -6	V12' +0.5	"	"

See footnote at end of table.

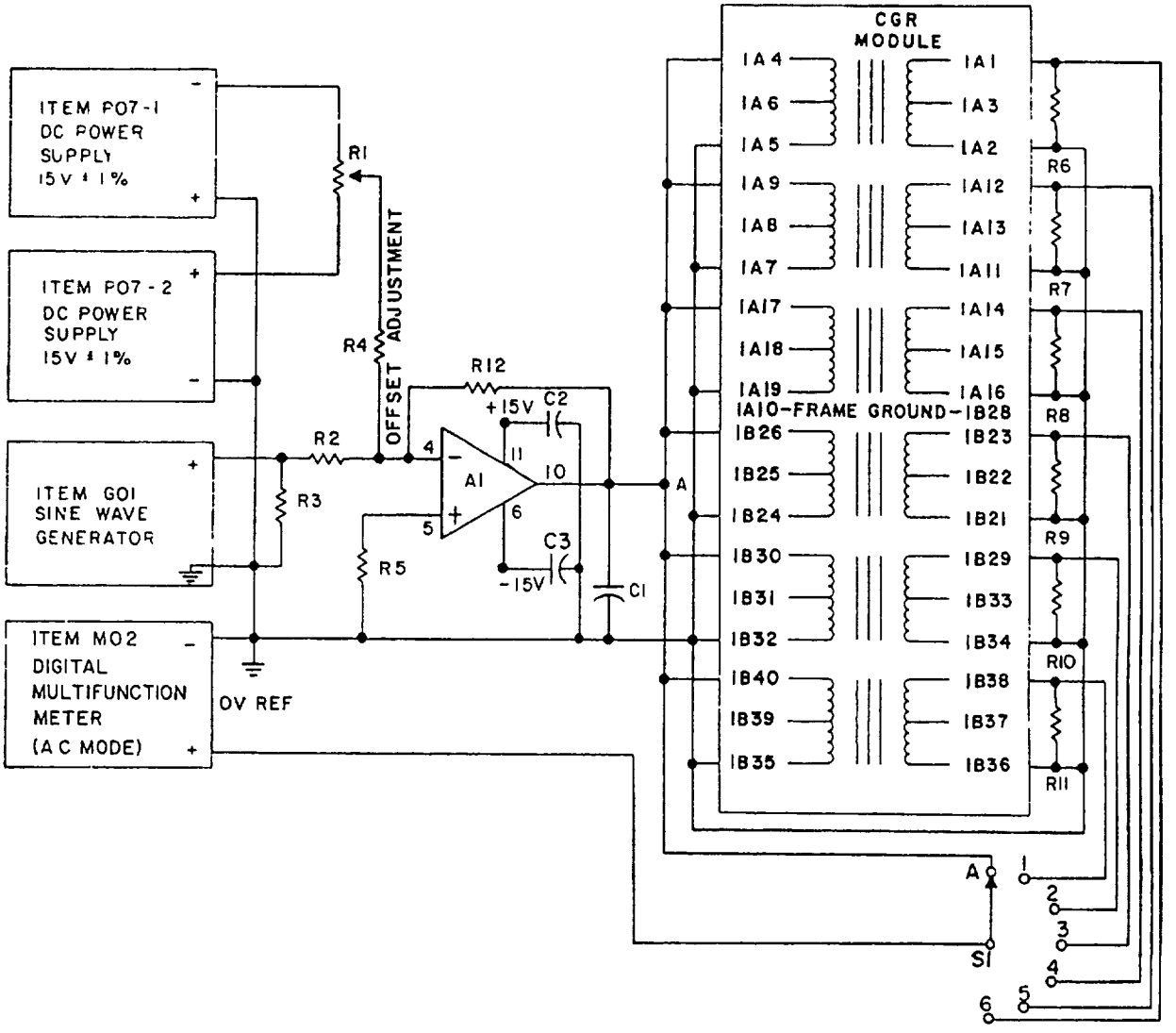
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* TABLE VIII. Frequency response check test - Continued.

Test condition	Parameter	Record item M02 reading 1/	S1 position	Item G01 frequency ±1%	Initial limits 25°C ±5°C ambient		Unit	Test figure
					Min	Max		
G37	Frequency response check	V7 =	1	5 kHz	0.174	0.202	Vrms	9
G38	"	V8 =	2	"	"	"	"	"
G39	"	V9 =	3	"	"	"	"	"
G40	"	V10 =	4	"	"	"	"	"
G41	"	V11 =	5	"	"	"	"	"
G42	"	V12 =	6	"	"	"	"	"
G43	"	- - -	1	300 kHz	V7' -4	V7' +0.5	dBV	"
G44	"	"	2	"	V8' -4	V8' +0.5	"	"
G45	"	"	3	"	V9' -4	V9' +0.5	"	"
G46	"	"	4	"	V10' -4	V10' +0.5	"	"
G47	"	"	5	"	V11' -4	V11' +0.5	"	"
G48	"	"	6	"	V12' -4	V12' +0.5	"	"

1/ If readings have previously been taken for test conditions C1-C6, they can be used for test conditions G1-G6 rather than repeating the test.

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* FIGURE 9. Insertion loss check and frequency response check test circuit for tables VII and VIII.

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NOTES:
1. R1 = 10 kΩ ±1%, multi-turn, 1/4 W, potentiometer.
2. R2 = R12 = 10 kΩ ±1%, 1/4 W.
3. R3 = 51Ω ±5%, 1 W.
4. R4 = 200 kΩ ±5%, 1/8 W.
5. R5 = 4.99 kΩ ±1%, 1/8 W.
6. R6-R11 = 2 kΩ ±1%, 1/8 W.
7. C1 = 300 pF ±10%.
8. C2 = C3 = 0.1 µF ±20%.
9. A1 = µA741 operational amplifier.
10. S1 = SP, 7 position, switch.
11. Adjust R1 for minimum offset out of A1 with item G01 and A1 disconnected from module under test. Use item M02 to measure A1 offset.
12. Test conditions F1-F12:

    a. Adjust item G01 for 1 Vrms ±0.1% output at point A.
    b. Adjust item G01 to 500 Hz ±1% for test conditions F1-F6 and to 10 kHz ±1% for test conditions F7-F12.
    c. Compare readings at S1 positions 1-6 to table VII limits.

13. Test conditions G1-G48:

    a. Adjust item G01 for 1 Vrms ±0.1% output at point A for test conditions G1-G30 and for 0.2 Vrms ±1% output at point A for test conditions G31-G48.
    b. Adjust item G01 frequency as specified in table VIII.
    c. Record in the blanks provided in table VIII the actual item M02 readings for V1-V6 and V7-V12.
    d. Convert V1-V6 and V7-V12 to dBV values V1'-V6' and V7'-V12'.
    e. Compare readings at S1 positions 1-6 to table VIII limits.
    f. For test conditions G7-G12 one or more inputs may have to be disconnected in order to obtain 1 Vrms at point A. Do not disconnect the input to the transformer under test.

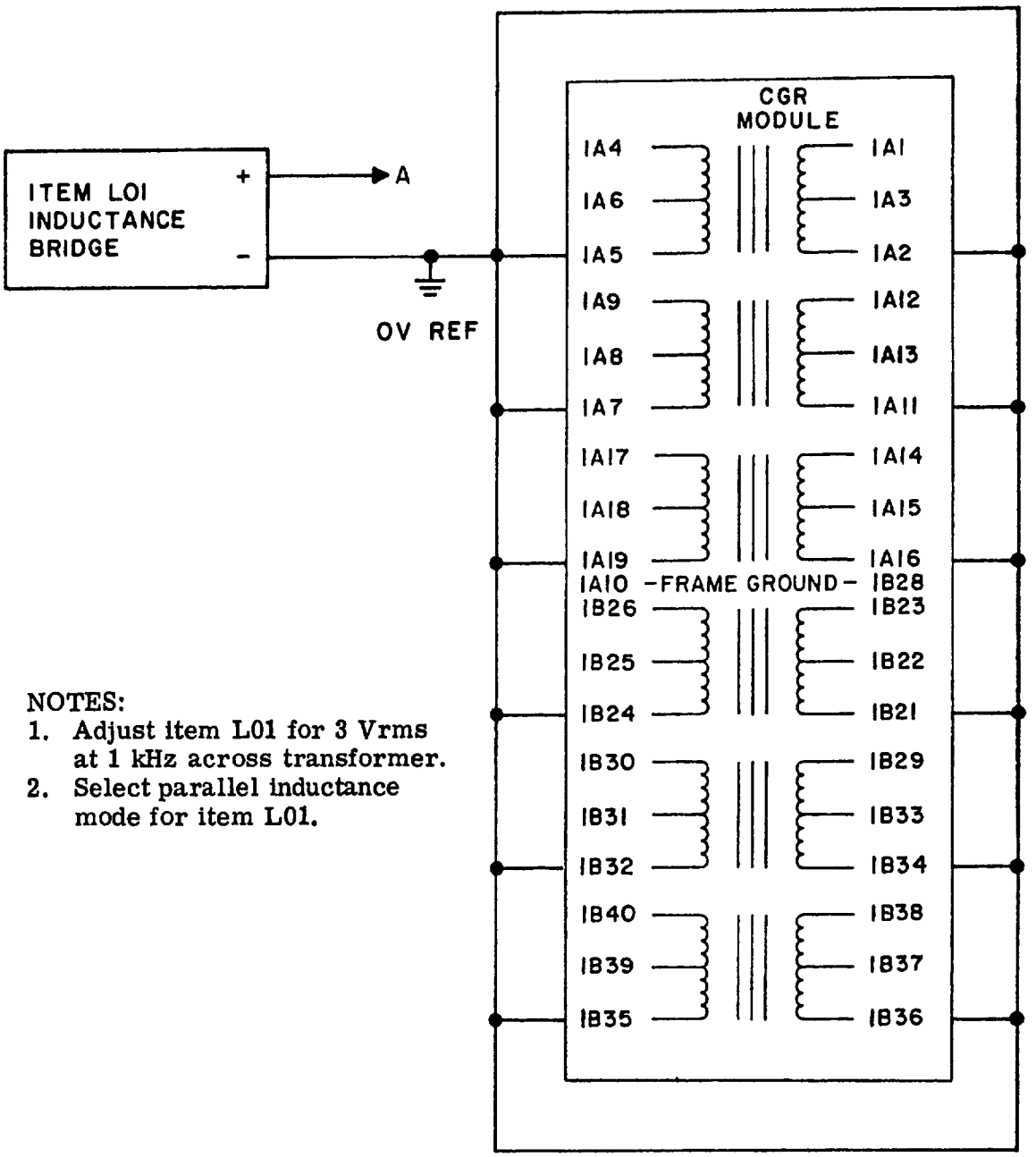
* FIGURE 9. Insertion loss check and frequency response check test circuit for tables VII and VIII - Continued.

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* TABLE IX. Primary inductance test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
H1	Primary inductance (1 kHz)	4	910	∞	mH	10
H2	"	9	"	"	"	"
H3	"	17	"	"	"	"
H4	"	26	"	"	"	"
H5	"	30	"	"	"	"
H6	"	40	"	"	"	"

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NOTES:
1. Adjust item L01 for 3 Vrms at 1 kHz across transformer.
2. Select parallel inductance mode for item L01.

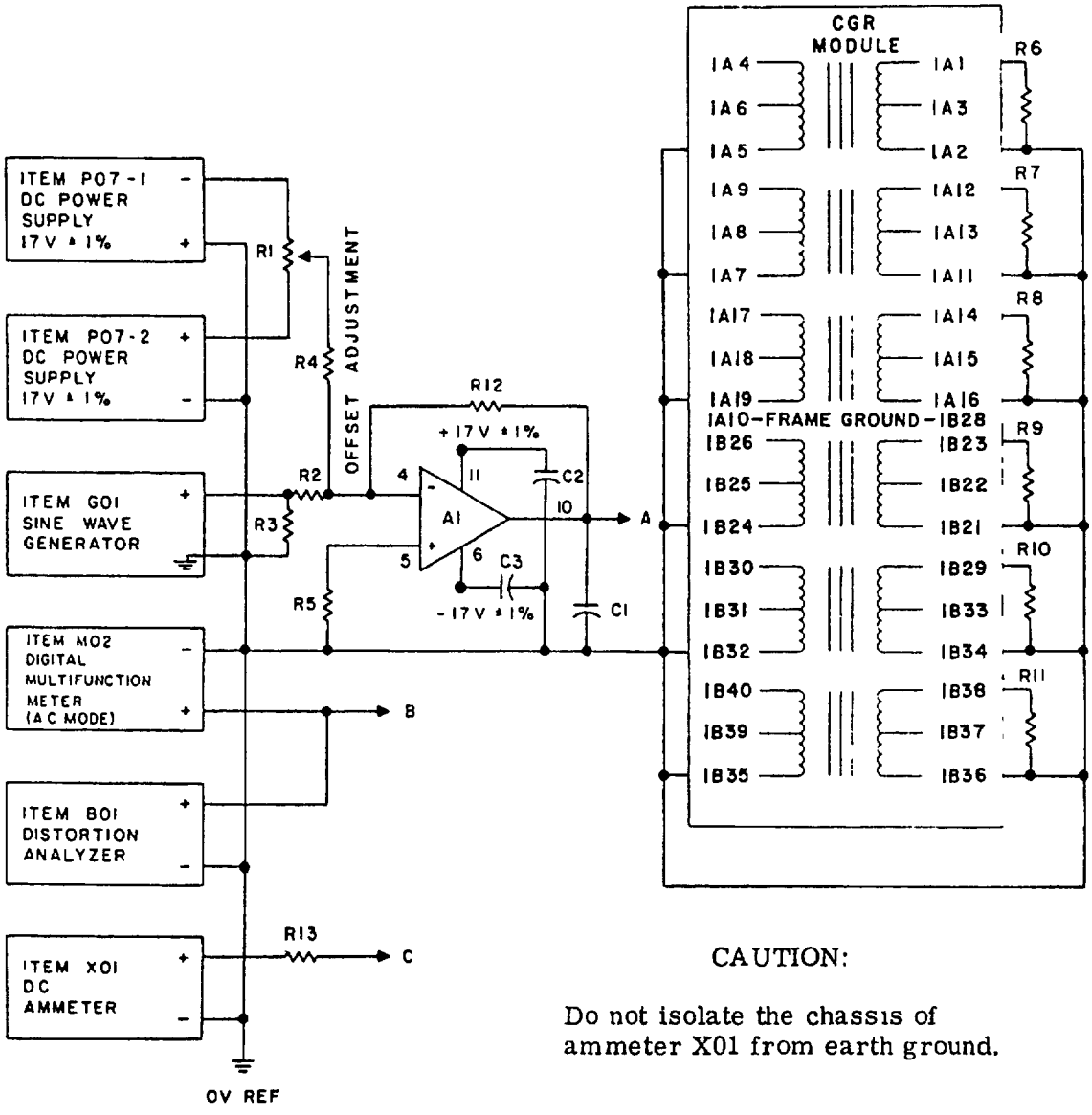
FIGURE 10. Primary inductance test circuit for table IX.

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TABLE X. Maximum power check (distortion test) test.

Test condition	Parameter	Item G01 frequency ±1%	Item M02 reading ±1%	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	Min	Max		
J1	Maximum power check (distortion test)	200 Hz	1 Vrms	4	1	0	5	%	11
J2	"	"	"	9	12	"	"	"	"
J3	"	"	"	17	14	"	"	"	"
J4	"	"	"	26	23	"	"	"	"
J5	"	"	"	30	29	"	"	"	"
J6	"	"	"	40	38	"	"	"	"
J7	"	400 Hz	4.5 Vrms	4	1	"	"	"	"
J8	"	"	"	9	12	"	"	"	"
J9	"	"	"	17	14	"	"	"	"
J10	"	"	"	26	23	"	"	"	"
J11	"	"	"	30	29	"	"	"	"
J12	"	"	"	40	38	"	"	"	"
J13	"	1 kHz	8.5 Vrms	4	1	"	"	"	"
J14	"	"	"	9	12	"	"	"	"
J15	"	"	"	17	14	"	"	"	"
J16	"	"	"	26	23	"	"	"	"
J17	"	"	"	30	29	"	"	"	"
J18	"	"	"	40	38	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 10 kΩ ±1%, multi-turn, 1/4 W potentiometer.
2. R2 = R12 = 10 kΩ ±1%, 1/4 W.
3. R3 = 600Ω ±5%, 1/4 W.
4. R4 = 200 kΩ ±5%, 1/8 W.
5. R5 = 4.99 kΩ ±1%, 1/8 W.
6. R6-R11 = 2 kΩ ±1%, 1/8 W.
7. R13 = 80Ω ±5%, 1/8 W.
8. C1 = 300 pF ±10%.
9. C2 = C3 = 0.1 µF ±20%.
10. A1 = µA741 operational amplifier.
11. With item G01 set at 0 Vrms output, connect A to C and adjust R1 for a current flow of 2.5 mA ±1% into item X01. Then disconnect A and C.
12. Connect A and B as specified in table X. Adjust item G01 for item M02 readings specified in table X. Using item B01 measure distortion D_T.
13. Measure distortion out of A1 at point A (D_A). Compute distortion (D_M) as follows

$$D_M = \sqrt{D_T^2 - D_A^2}.$$

Compare D_M to limits specified in table X.

* FIGURE 11. Maximum power check (distortion test) test circuit for table X.

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* TABLE XI. Magnetic shielding check test.

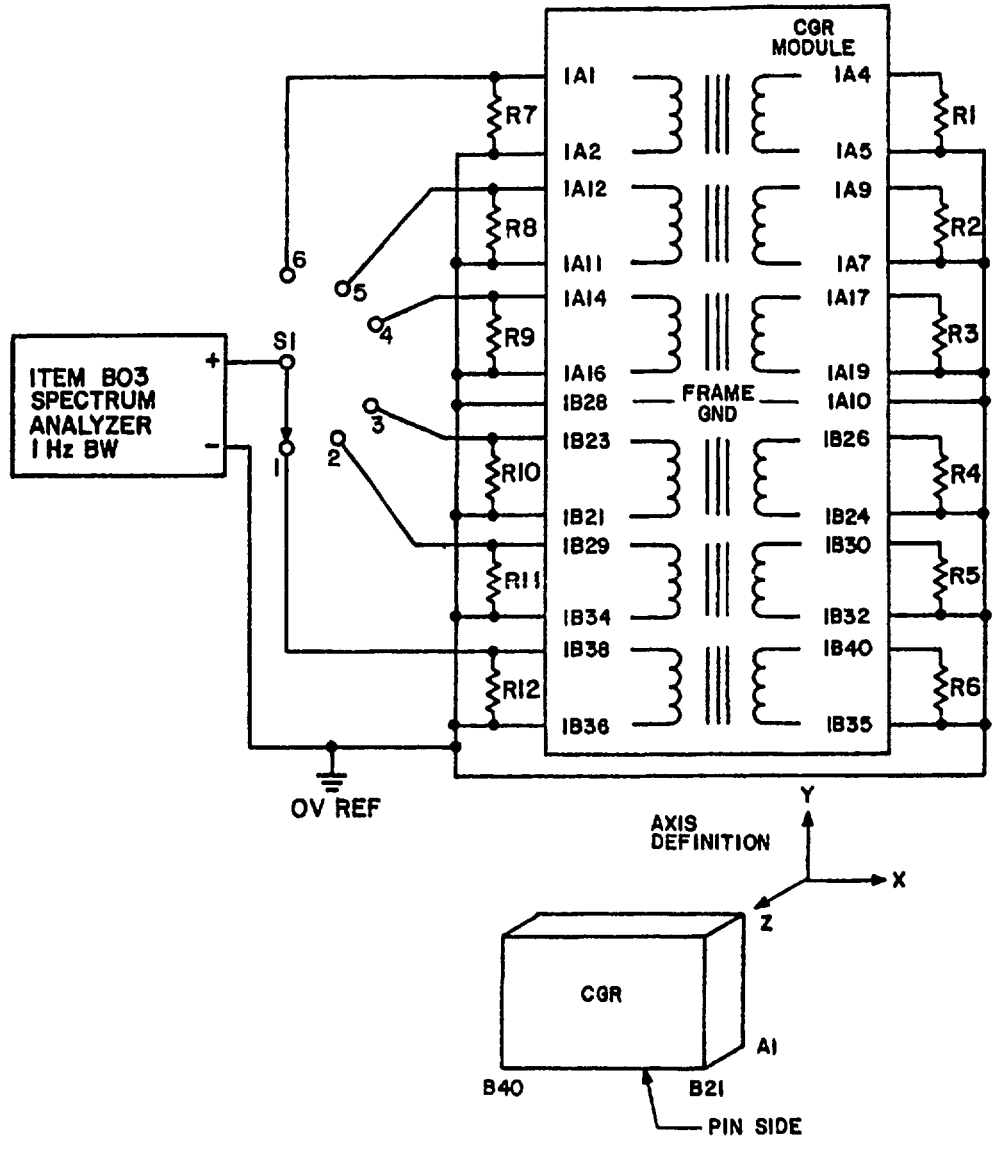
Test condition	Parameter	Module orientation	S1 position	Helmholtz coil frequency ±1%	Initial limits 25°C ±5°C ambient		Unit	Test figure
					Min	Max		
K1	Magnetic shielding check	X-axis	1	60 Hz	-∞	-80	dBV	12
K2	"	"	2	"	"	"	"	"
K3	"	"	3	"	"	"	"	"
K4	"	"	4	"	"	"	"	"
K5	"	"	5	"	"	"	"	"
K6	"	"	6	"	"	"	"	"
K7	"	"	1	400 Hz	"	"	"	"
K8	"	"	2	"	"	"	"	"
K9	"	"	3	"	"	"	"	"
K10	"	"	4	"	"	"	"	"
K11	"	"	5	"	"	"	"	"
K12	"	"	6	"	"	"	"	"
K13	"	Y-axis	1	60 Hz	"	"	"	"
K14	"	"	2	"	"	"	"	"
K15	"	"	3	"	"	"	"	"
K16	"	"	4	"	"	"	"	"
K17	"	"	5	"	"	"	"	"
K18	"	"	6	"	"	"	"	"

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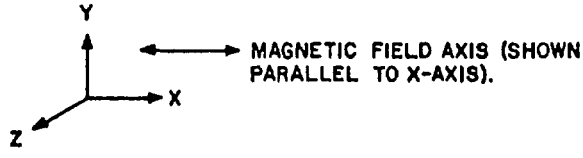
* TABLE XI. Magnetic shielding check test - Continued.

Test condition	Parameter	Module orientation	S1 position	Helmholtz coil frequency ±1%	Initial limits 25°C ±5°C ambient		Unit	Test figure
					Min	Max		
K19	Magnetic shielding check	Y-axis	1	400 Hz	-∞	-80	dBV	12
K20	"	"	2	"	"	"	"	"
K21	"	"	3	"	"	"	"	"
K22	"	"	4	"	"	"	"	"
K23	"	"	5	"	"	"	"	"
K24	"	"	6	"	"	"	"	"
K25	"	Z-axis	1	60 Hz	"	"	"	"
K26	"	"	2	"	"	"	"	"
K27	"	"	3	"	"	"	"	"
K28	"	"	4	"	"	"	"	"
K29	"	"	5	"	"	"	"	"
K30	"	"	6	"	"	"	"	"
K31	"	"	1	400 Hz	"	"	"	"
K32	"	"	2	"	"	"	"	"
K33	"	"	3	"	"	"	"	"
K34	"	"	4	"	"	"	"	"
K35	"	"	5	"	"	"	"	"
K36	"	"	6	"	"	"	"	"

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NOTES:
1. R1-R6 = 51Ω ±5%, 1/8 W.
2. R7-R12 = 2 kΩ ±1%, 1/8 W.
3. S1 = SP, 6 position, switch.
4. All signal leads shall be twisted-shielded pairs.
5. Magnetic field generation shall be by the use of a Helmholtz coil. Coil description and module positioning shall comply with MIL-T-27.
6. Adjust the excitation to the coil for a magnetic field of 1 Gauss ±10% at 60 Hz or 400 Hz per table XI. Orient the module in the field per table XI where field is oriented as shown:



7. With coil energized, make item B03 readings at S1 positions 1-6 at 60 Hz and 400 Hz per table XI, Note 7 readings shall not exceed table XI limits.

FIGURE 12. Magnetic shielding check test circuit for table XI.

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TABLE XII. Output crosstalk test.

Test condition	Parameter	Items G01 and B03 frequency	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	Output crosstalk	500 Hz	4	12	-∞	-70	dBV	13
L2	"	"	"	14	"	"	"	"
L3	"	"	"	23	"	"	"	"
L4	"	"	"	29	"	"	"	"
L5	"	"	"	38	"	"	"	"
L6	"	"	9	14	"	"	"	"
L7	"	"	"	23	"	"	"	"
L8	"	"	"	29	"	"	"	"
L9	"	"	"	38	"	"	"	"
L10	"	"	17	23	"	"	"	"
L11	"	"	"	29	"	"	"	"
L12	"	"	"	38	"	"	"	"
L13	"	"	26	29	"	"	"	"
L14	"	"	26	38	"	"	"	"
L15	"	"	30	38	"	"	"	"
L16	"	5 kHz	4	12	"	"	"	"
L17	"	"	"	14	"	"	"	"
L18	"	"	"	23	"	"	"	"
L19	"	"	"	29	"	"	"	"

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TABLE XII. Output crosstalk test - Continued.

Test condition	Parameter	Items GO1 and BO3 frequency	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L20	Output crosstalk	5 kHz	4	38	-∞	-70	dBV	13
L21	"	"	9	14	"	"	"	"
L22	"	"	"	23	"	"	"	"
L23	"	"	"	29	"	"	"	"
L24	"	"	"	38	"	"	"	"
L25	"	"	17	23	"	"	"	"
L26	"	"	"	29	"	"	"	"
L27	"	"	"	38	"	"	"	"
L28	"	"	26	29	"	"	"	"
L29	"	"	26	38	"	"	"	"
L30	"	"	30	38	"	"	"	"
L31	"	10 kHz	4	12	"	"	"	"
L32	"	"	"	14	"	"	"	"
L33	"	"	"	23	"	"	"	"
L34	"	"	"	29	"	"	"	"
L35	"	"	"	38	"	"	"	"
L36	"	"	9	14	"	"	"	"
L37	"	"	"	23	"	"	"	"
L38	"	"	"	29	"	"	"	"

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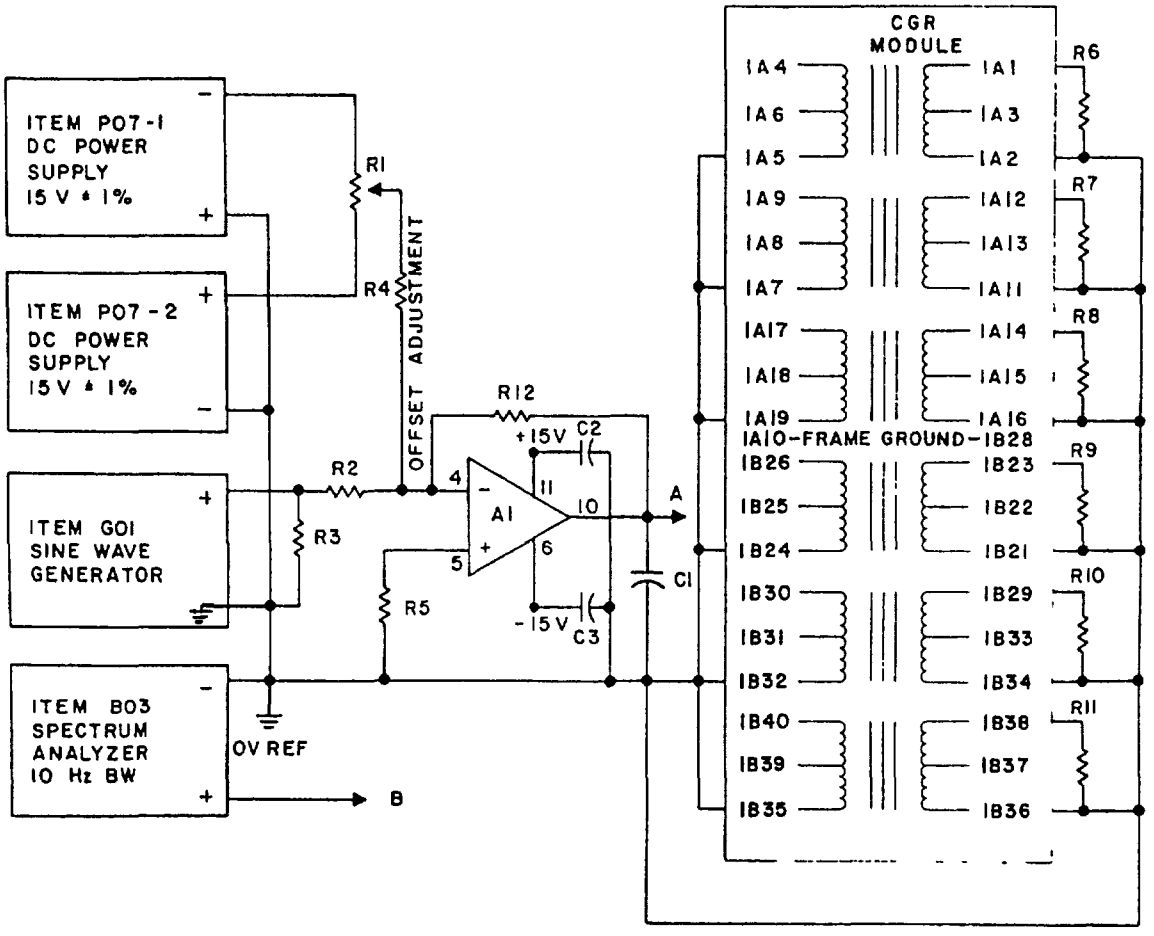
TABLE XII. Output crosstalk test - Continued.

Test condition	Parameter	Items G01 and B03 frequency	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L39	Output crosstalk	10 kHz	9	38	-∞	-70	dBV	13
L40	"	"	17	23	"	"	"	"
L41	"	"	"	29	"	"	"	"
L42	"	"	"	38	"	"	"	"
L43	"	"	26	29	"	"	"	"
L44	"	"	26	38	"	"	"	"
L45	"	"	30	38	"	"	"	"

TABLE XIII. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Continuity	10	28	0	1	Ω	14
M2	Continuity	10	Frame	0	1	Ω	14

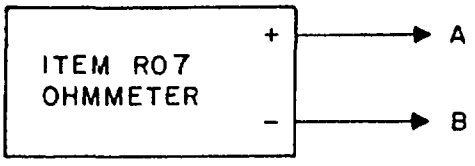
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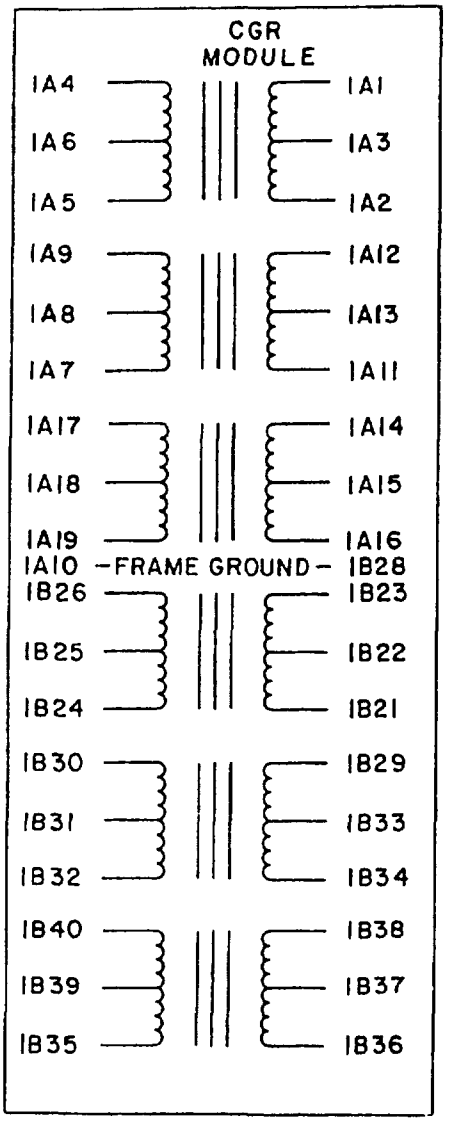


NOTES:
1. R1 = 10 kΩ ±1%, multi-turn, 1/4 W, potentiometer.
2. R2 = R12 = 10 kΩ ±1%, 1/4 W.
3. R3 = 51Ω ±5%, 1 W.
4. R4 = 200 kΩ ±5%, 1/8 W.
5. R5 = 4.99 kΩ ±1%, 1/8 W.
6. R6-R11 = 2 kΩ ±1%, 1/8 W.
7. C1 = 300 pF ±10%.
8. C2 = C3 = 0.1 µF ±20%.
9. A1 = µA741 operational amplifier.
10. Adjust R1 for minimum offset out of A1 with item G01 and A1 disconnected from module under test. Use item M02 to measure A1 offset.
11. Adjust item G01 for 1 Vrms ±5% output at point A using item B03.
12. Adjust items G01 and B03 to 500 Hz ±0.1% for test conditions L1-L15, to 5 kHz ±0.1% for test conditions L16-L20; and to 10 kHz ±0.1% for test conditions L31-L45.
13. Prior to performing test conditions at each frequency, connect A to B and adjust item B03 frequency for maximum reading.

* FIGURE 13. Output crosstalk test circuit for table XII.

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NOTE: On test condition M2 (when measuring to frame), penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 14. Continuity test circuit for table XIII.

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